

UNIT II

INTRODUCTION TO JOINING PROCESSES

Weldability

Weldability of a material is defined as its ability to be welded. Weldability of a material depends on various factors such as metallurgical changes that occur during the process of welding, its harness, gas evolution and absorption, extent of oxidation and others, Low carbon steels ($C \leq 0.12\%$) have the best weldability amongst metals.

As the carbon content increases, weldability decreases.

Selection of welding processes on the basis of materials being joined

<i>Materials to be welded</i>	<i>Recommended joining processes</i>
• Plain carbon steel and low-alloy steel • Cast iron • Stainless Steel	Arc welding, Oxyacetylene welding, Resistance welding and Brazing (best results); EBW and Adhesive bonding (commonly performed) Oxyacetylene welding (best results); Arc welding and Adhesive bonding (commonly performed) Arc welding, Resistance welding and Brazing (best result); Oxyacetylene welding, EBW, Soldering and Adhesive bonding (commonly performed)
• Aluminum and Magnesium • Copper and Copper alloys • Thermoplastics and thermosets • Ceramics • Dissimilar metals	Adhesive bonding (best results); Arc welding, Oxyacetylene welding, EBW, Resistance welding and Brazing (commonly performed) Brazing and Soldering (best result); Arc welding, Oxyacetylene welding, EBW, Resistance welding and Adhesive bonding (commonly performed) Adhesive bonding (commonly performed) Adhesive bonding (best results); EBW (commonly performed) Soldering and Adhesive bonding (best results); EBW (commonly performed)

Selection of welding processes on the basis of material thickness

<i>Materials thickness</i>	<i>Recommended joining processes</i>
• Very thin category($\leq 3\text{mm}$)	Resistance welding, OFG, GMAW, GTAW, FCAW, LBW, Ultrasonic welding and EBW (low power)
• Thin category (3-6 mm) and thick category (6-20 mm)	GMAW, SAW, FCAW, LBW low (high power) and EBW (medium power); Multirun weld can be used if necessary.
• High thick category (20-75mm) and very high thick category ($>75\text{mm}$)	SAW, ESW, EBW (high power) and termite welding

Selection of welding processes on the basis of rate of heat input

<i>Materials thickness</i>	<i>Recommended joining processes</i>
• Low • Moderate • High	OFG, ESW and Resistance welding (flash SMAW, FCAW, GMAW, SAW and GTAW PAW, EBW, LBW and Resistance welding (spot, seam and percussion)

Selection of welding processes on the basis of types of joint

<i>Types of joint</i>	<i>Recommended joining processes</i>
• Lap joint	OFG(acetylene), SMAW, GTAW, PAW, GTAW, FCAW, Resistance welding (spot and seam) and EBW
• Butt joint (bars)	Resistance welding (butt and flash), Friction welding , EBW, Thermit welding and Diffusion bonding
• Butt joint (tubes)	SMAW, GTAW, PAW, GMAW, FCAW, Resistance welding (butt and flash), Friction welding, EBW and Diffusion bonding
• T-joint or fillet	SMAW, SAW, GTAW, GMAW, FCAW, ESW, EGW, Resistance welding (projection), EBW, Thermit welding and Diffusion bonding

Selection of welding processes on the basis of welding positions

<i>Welding positions</i>	<i>Recommended joining processes</i>
• Flat or down hand	SAW, GMAW and FCAW
• Horizontal (fillet)	SAW, GMAW and FCAW
• Horizontal	GMAW and FCAW
• Vertical	GMAW, FCAW and ESW
• Overhead	GMAW and FCAW

Selection of welding processes on the basis of deposition rate

<i>Deposition</i>	<i>Recommended joining processes</i>
• Low • Medium • High • Very high	SMAW and GTAW GMAW GMAW and FCAW SAW and ESW(both using iron powder coated electrodes)

Selection of welding processes on the basis of economic consideration

<i>Deposition</i>	<i>Recommended joining processes</i>
• Low cost • Medium cost • High cost	SMAW, ESW GTAW, GMAW, stud welding, Resistance welding, OFW and Thermit welding SAW, ESW and Diffusion bonding EBW, LBW and Friction welding

Selection of welding processes on the basis of shape of job

<i>Deposition</i>	<i>Recommended joining processes</i>
• Sheet and plate • Large pipe and cylinder	OFW (acetylene), SMAW, SAW, GTAW, PAW, GMAW, FCAW, ESW, EGW, Resistance welding(spot and seam) and EBW OFW (acetylene), SMAW, SAW, GTAW, PAW, GMAW, FCAW, Resistance welding(spot and seam) and EBW

WELD DEFECTS

The study of defects in a welded joint is necessary to find out the reason of failure because the defective weldment may fail and cause damage during its service life.

Defects reduce the strength of the joint and may arise due to the use of sub-standard materials, defective welding equipment or poor welding skill.

The major defects or discontinuities that affect weld quality are discussed below.

Porosity

Porosities are defined as voids, holes or cavities usually of spherical shapes. It is caused by gas entrapped in the weld metal during solidification .additionally, chemical reactions during welding.

Porosity in welds may be of the following types:

- Uniformly scattered porosity
- Cluster porosity
- Starting porosity
- Liner porosity

Shrinkage Cavity

Shrinkage cavity normally forms during the welding of thick plates by arc welding processes especially those dealing with large amount of molten metal. This defect occurs at the surface and may extend inside the weld and is formed due to shrinkage of weld metal during its solidification.

Cracks

Cracks are the most dangerous of all weld defects. They may occur in various locations and directions in the weld area such as base metal, base and weld metal interface or weld metal on its surface, under the weld bead or in the crater. *Transverse, longitudinal* and *crater cracks* are defines as surface cracks because they are formed on the surface of the weld.

Slag inclusion

Inclusions in the form of slag such as oxides, suiphides, fluxes and other foreign material may get entrapped in the weld metal and base metal, and do not get change to escape and hence float on the surface. Inclusions lower the strength of the joint.

Distortion

Distortion may result from the differential rate of heating and cooling in the weld zone or adjacent metal leading to the generation of stresses. Parts after joining may not look properly aligned due to distortion. Reducing welding current, using small diameter electrode and proper clamping devices can reduce or even prevent distortion completely.

Incomplete Fusion and Penetration

Incomplete fusion often result from poor join preparation, wrong joint design or incorrect weld parameters such as welding current, welding speed etc. However, incomplete or poor penetration results from too little heat input due to decrease in welding current.

Incorrect Weld Profile

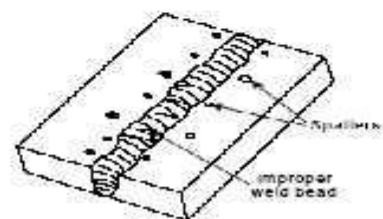
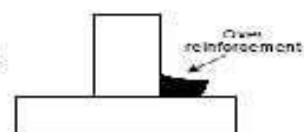
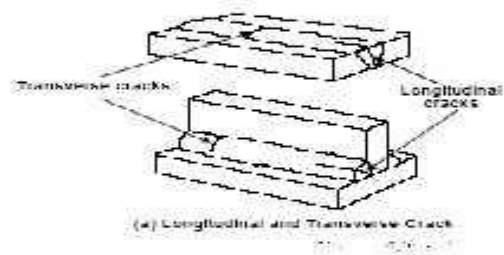
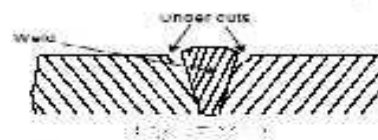
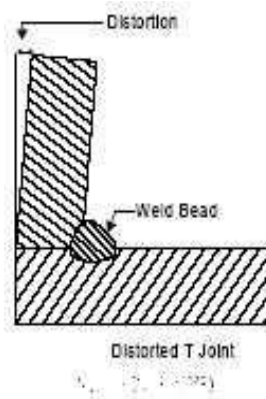
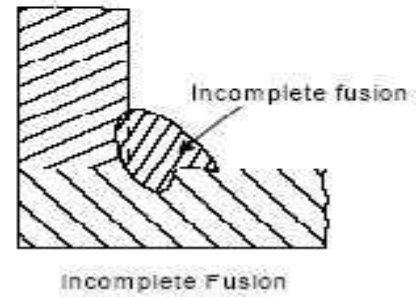
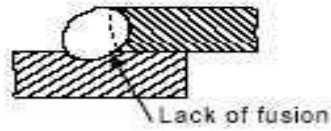
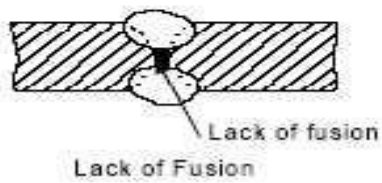
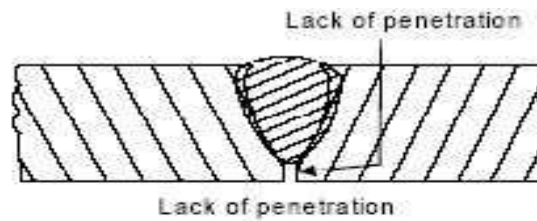
Wrong bead profile can be a source for lack of strength and bad appearance of the joint. Excessive reinforcement or excessive penetration may be due to bad weld profile and can be a source of stress concentration.

Undercutting and Overlapping

In undercutting, a groove gets formed in the base metal along the sides of the weld bead. Due to excessive welding current and are voltage, excessive melting of fusion faces of the base metal takes place and a sharp recess or notch is formed.

Spatter

Spatters are the small metal particles that are thrown randomly in all directions around the arc and are deposited on the base metal. It degrades the quality of the surface around the joint. Excessive spatter occurs due to high welding current, are blow and wrong selection of electrodes.



Poor Weld Bead Appearance

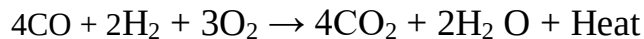
GAS WELDING PROCESSES

Oxyacetylene Welding

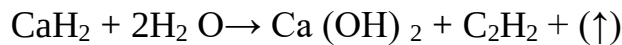
Oxyacetylene Welding use acetylene-oxygen mixture to the required heat. The chemical reaction between acetylene and oxygen is exothermic in nature and sufficient heat is released at the end of the reaction which has a temperature of around 3250°C . This heat is use to melt the parts to be joined. The reaction takes place in tow stages. In the first stage, acetylene and oxygen react to produce carbon monoxide and hydrogen. Some heat is also liberated during the process.



In the second stage, the products formed in the first stage further react in presence of excess oxygen production about tow –thirds of the total heat.



Acetylene gas used in the process is obtained when calcium carbide (CaC_2) chemically reacts with water.



Acetylene forms explosive mixtures with air is not safe to store it as a gas at pressures higher than 0.1 MPa. Hence, it is usually dissolved in acetone which has the property of absorbing 25 times its own volume of acetylene for each atmosphere of pressure applied.

Advantages

- The equipment cost is low and requires little maintenance.
- It is portable and can be used anywhere.
- The gas flame is generally more easily controlled.
- The process can also be used for cutting

Limitations

- Since the heat source is not concentrated, a large area of the metal is heated and distortion is likely to occur.
- In some cases, there is a loss of corrosion resistance.
- The process is very slow. Therefore, it has been largely replaced by arc welding.
- Proper operator training and skill are also essential.

Suitability and Applications

Oxyfuel gas welding is used extensively for sheet- metal fabrication and repair works. With proper technique, practically all metals can be welded. Quality of the weld depends upon the way how the surface are made clean and prevented from atmospheric contamination.

Welding Equipment for Oxyacetylene Welding

1. Gas Cylinders

Pressure

Oxygen – 125 kg/cm^2

Acetylene – 16 kg/cm^2

2. Regulators

Working pressure of oxygen 1 kg/cm²

Working pressure of acetylene 0.15 kg/cm²

Working pressure varies depends upon the thickness of the work pieces welded.

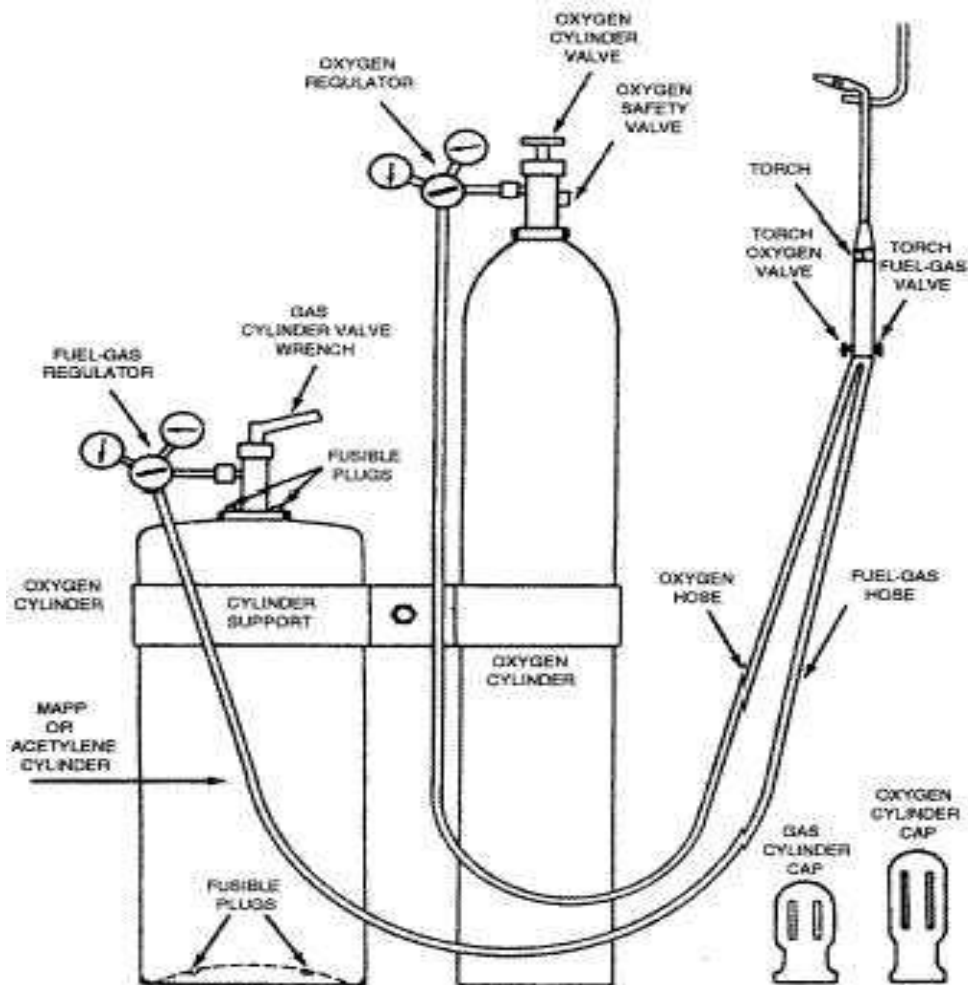
3. Pressure Gauges

4. Hoses

5. Welding torch

6. Check valve

7. Non return valve



The equipment for oxyacetylene gas welding consists of a welding torch (blowpipe) which is connected by hoses to high pressure gas cylinders and equipped with pressure gauges and regulators.

Oxygen regulators are usually painted green, and acetylene regulators red. Other equipment includes safety equipments such as goggles with shaded lenses, face shields, gloves and protective clothing.

Types of Flames used in Oxyacetylene Welding

Three distinct flame variations are produced with oxyacetylene gas mixture namely, reducing or carburizing, neutral and oxidizing. The correct adjustment of the flame is important for reliable welds.

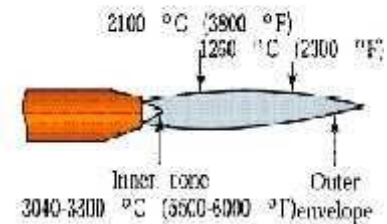
If acetylene and oxygen are present in equal proportions (ratio is 1:1), neutral flame is produced. For most welding operations, a neutral flame is used, since it has least chemical effect on the heated metal.

If the flame has high oxygen content (the ratio of oxygen and acetylene is 1.5:1), then it is known as oxidizing flame. Such flames are used in the welding of copper and copper alloys (brass and bronze). This flame is harmful, especially for steels, because it oxidizes the steel.

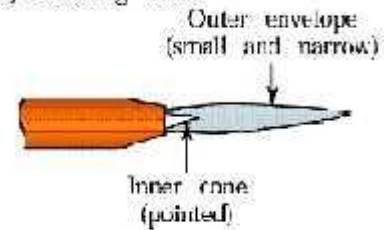
If the flame contains excess of acetylene, it is known as carburizing flame. The excess fuel decomposes to carbon and hydrogen in the absence of sufficient oxygen and the temperature produced by such flame is lower (3050°C). Hence, it is suitable for application requiring low heat, such as brazing, soldering and flame hardening.

Carburizing flame is used in the welding of Monel metal (a nickel – copper alloy), high –carbon steel, nickel, certain alloy steels, and many of the non-ferrous, hard – surfacing materials. Steel is not welded by using this flame since it will increase the carbon content in the weld resulting in a hard and brittle weld.

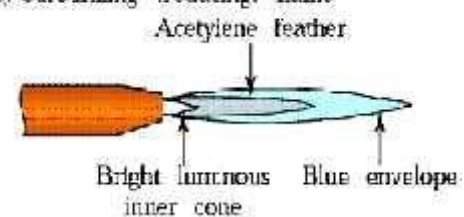
(a) Neutral flame



(b) Oxidizing flame



(c) Carburizing (reducing) flame



ARC WELDING PROCESSES

The various arc welding processes are:

1. Carbon Arc Welding
2. Shielded Metal Arc Welding
3. Flux Cored Arc Welding
4. Gas Tungsten Arc Welding
5. Gas Metal Arc Welding
6. Plasma Arc Welding
7. Atomic Hydrogen Welding
8. Electro slag Welding
9. Stud Arc Welding
10. Electro gas Welding

Shielded Metal Arc Welding (SMAW)

Shielded metal arc welding is the most widely used arc welding process using consumable electrode. It is one of the oldest, simplest and most versatile joining processes. It is also known as stick welding, because of stick like shape of the electrodes used in the process. The heat is produced by an arc between the flux- covered electrode and the work piece. The

process is shown in The heat of the arc melts the tip of the electrode, its coating and the work piece just below the arc. The electrode coating, on melting, forms shielding gas which protects the molten metal from atmospheric contamination. It also stabilizes the arc. The flux used in the process combines with the impurities present in the molten metal and floats on its surface in the form of slag. It is chipped away from the surface on cooling.

Advantages

The SMAW equipment is simple, portable and less expensive compared to other arc- welding equipments. Welds can be made in all positions.

Limitation

The process used finite length electrode which gets consumed and it is required to use new electrodes frequently. This makes process slower. Because some of the heat is taken by the electrode, its quality deteriorates with time. Slag if left unremoved causes corrosion resulting in the failure of the joint. Low melting point metals such as zinc, lead and tin arc not welded by SMAW because of burn-off possibility.

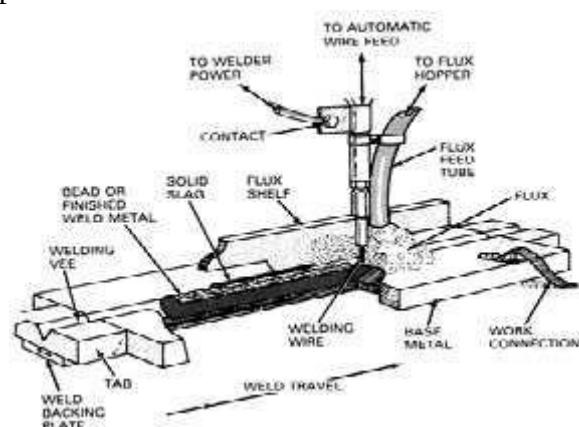
Applications

SMAW is used extensively by many of the fabrication industries such as general construction, shipbuilding, pipelines and pressure vessel fabrication. It is also used in repair operations. Carbon steels, alloy steels, stainless steel and cast irons arc commonly welded metals.

Submerged Arc Welding (SAW)

In the submerged arc welding, arc is produced between a consumable metal electrode and the work piece. Since the arc is submerged (hidden) under a heavy coating of granular flux, hence the name submerged arc welding. The arc travels beneath the surface of flux mass fed from a tube.

The flux consists of calcium or magnesium silicate. A portion of the flux metal and combines with the impurities to form the slag to be finally chipped away on cooling. Slag if left unremoved causes corrosion resulting in the failure of the joint. The remaining unmelted flux provides shielding atmosphere.



Copper coated steel wire is used as electrode. The large diameter electrodes are used because of their high current carrying capacity and fast deposition rate. The process can be performed manually or it may be semiautomatic or fully automatic of which the latter is more common today. The process of submerged arc welding .

Both alternating and direct current can be used in this process. Direct current reverse polarity (dcrp) gives deeper penetration. Direct current straight polarity (dcsp) ensures higher deposition rate.

Iron powder can be applied directly into the joint to increase the metal deposition rate which can make the process very fast. The resulting process is called bulk welding.

Advantages

- Since the arc is completely hidden, there is no chance of any flash, spatter or smoke visible from outside.
- Heavy currents can be used. In other arc welding processes, heavier current are not preferred because of higher infrared and ultraviolet radiations. No such problem arises in this process because the arc is covered by the flux.
- Higher current gives high deposition rate and deep penetration. The deposition rate can be much more than shielded metal arc welding (SMAW) process.
- Welding speed is higher which helps to join thicker plates.
- It produces very high quality weld because of high degree of cleanliness.

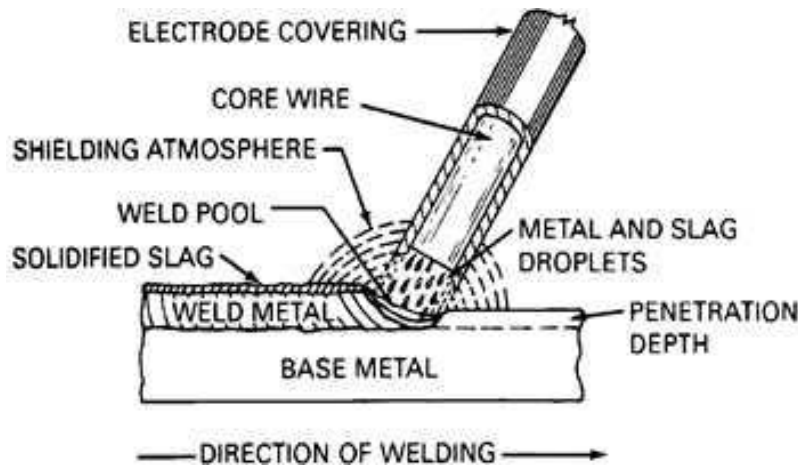
Limitations

- It is largely confined to flat position welding. Overhead welding is not possible because of the possibility of falling down of large molten pool and flux.
- It is not suitable for thinner metals because of burn – off possibility.
- The arc is difficult to guide.
- Due to huge amount of flux being used in the process, it has slow rate of cooling.

Shielded metal arc welding process (SMAW)

Shielded Metal Arc Welding (SMAW) Process is a manual welding process where an electric arc is used for localized melting of workpiece materials. The electric arc is created between a consumable electrode and the workpiece material as shown .

The electrode is covered with a coating (referred to as *flux*), which is extruded on the surface of the electrode. During welding, the electrode coating decomposes and melts providing a protective atmosphere on the weld area to avoid the reaction between hot liquid metal and atmospheric gases. SMAW process is the most popular amongst all other fusion arc welding processes since the equipment is inexpensive and easy to use. SMAW process can be used on carbon steels, low alloy steels, stainless steels, cast irons, copper, nickel, and aluminium.



Advantages

- The filler metal is transferred through the protected arc and there is no frequent change of electrodes as in the SMAW process. These two factors help GMAW to produce high-quality welds at high welding speeds. Welding productivity is double that of the SMAW process.
- No flux is required, and no slag forms over the weld. This makes the process cleaner.
- GMAW has excellent penetration; it produces sound welds at high speeds. this makes the process especially adapted to automatic operation and the process lends itself readily to robotics and flexible manufacturing systems(FMS).
- Welding can be performed in all positions.
- It is a very versatile process and can be used on both light and heavy-gage structural plates.

Limitations

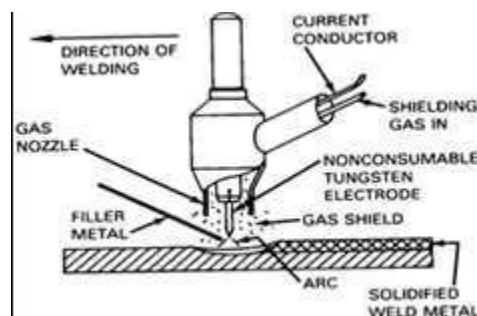
The cost of equipment and consumables is much higher as compared to SMAW process.

Gas Tungsten Arc Welding

Gas tungsten arc welding (GTAW) was formerly known as TIG (tungsten inert gas) welding. In this process, the arc is produced between a nonconsumable tungsten electrode and the work piece. Inert gases such as argon, helium or a mixture of the tow are employed to provide a protective shield around both the arc and the molten metal pool.

Selection of the current depends upon the type of material to be welded. DCSP gives deep penetration and faster welding on thicker work pieces and is used for welding steel, cast iron, copper alloys stainless steel.

DCRP provides a wide weld pool with shallow penetration and hence is ideal for thin workpieces, but it is rarely used because it tends to melt the tungsten electrode. In general, ac is preferred for aluminum and magnesium because the cleaning action of ac removes oxides and improves weld quality.



Advantages

- It gives a clean weld joint because of no use of flux in the process.
- There is less chance of weld spatter which makes the joint defect-free.
- It can be easily automated.
- Welding can be done in all positions.

Limitations

- The high cost of the inert gas makes this process more expensive than shielded metal arc welding (SMAW).
- The process is relatively slow as compared to GMAW and hence is not suitable for welding heavier metals.

Atomic Hydrogen Arc Welding

This welding process differs from other arc welding Processes in the sense that the arc is produced between two nonconsumable tungsten electrodes rather than between one electrode and the work piece. Hence, the work piece is out of the electrical circuit. Hydrogen gas is passed through the arc. The heat of the arc breaks the hydrogen molecules into atoms.

These hydrogen atoms after passing through the arc recombine to form molecules. This is a fusion reaction releasing large amount of heat having temperature of the order of 6100°C . This heat is utilized for joining the metals. Here heat of the arc is not directly used for welding purpose, rather it is used to break the molecular hydrogen into atomic hydrogen. Normally an alternating current is used because of its better current regulation.

Advantages

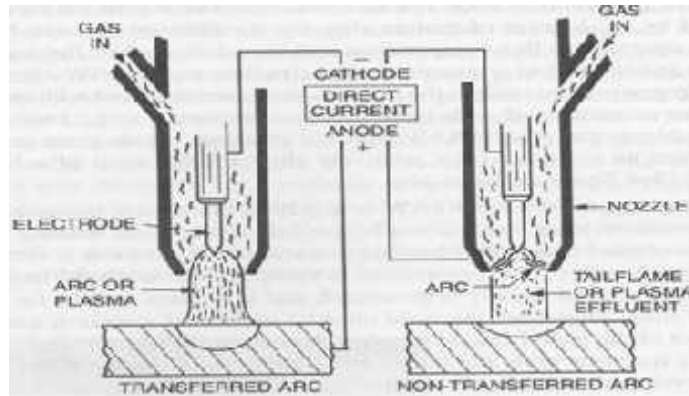
- It gives high heat concentration.
- There is no need for additional shield because hydrogen protects the electrodes and molten metal from oxidation.
- Due to high heat capacity, it can be used to used to join many difficult -to- weld alloy.

Suitability and Applications

It is an order welding method and has limited use today. It is mostly used for the repairing of steel moulds and dies.

Plasma arc welding process (PAW)

The objective of the Plasma Arc Welding (PAW) process is to increase the intensity of the of the arc plasma in a controlled manner such that greater thickness can be welded with the minimum spread of the welding arc. This objective is achieved by providing a special gas nozzle around a tungsten electrode operating on direct current electrode negative (DCEN) polarity. The constricted plasma formed is highly ionized and concentrated.

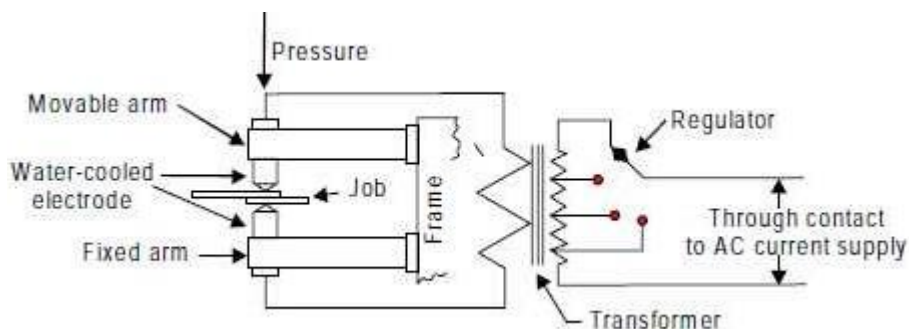


Two variants of the Plasma Arc Welding (PAW) process are commonly used. One is referred to as the *transferred arc process* and the second is referred to as the *non-transferred arc process*. In the *transferred arc process*, the workpiece is connected to the negative polarity so that the arc plasma irradiated on the workpiece has greater intensity

RESISTANCE WELDING PROCESSES.

Resistance Spot Welding

It is the simplest and most widely used method of resistance welding. A pair of copper electrodes is used to press the work pieces kept between them. A very small area, known as spot, is welded at the point where the workpiece are pressed by the electrodes. The metal at the spot melts and forms nugget. The weld nugget is generally 1.5 – 13 mm in diameter. The process is shown in Figure 12.2. Very high currents are used in the process. Selection of a particular value of current depends on the materials being welded and their thickness. The process uses no filler metal. It produces a clean weld because the joint is fully protected and there is no chance of atmospheric contamination. The welding cycle for spot welding is divided into four parts.



Squeeze time

During squeeze time, the work pieces are pressed between the electrodes to make intimate contact. No current flows through the work pieces during this time.

Weld time

During this period, the current is switched on while maintaining the maximum squeeze pressure. The temperature at the interface starts rising and attains welding temperature to melt the metal at the interface. The pressure is now increased considerably. Pressure plays an important role in resistance welding. The resistance at the interface is inversely proportional to the pressure. The variation is shown in Figure 12.3. If the pressure is low, the contact resistance will be high

which can burn the surface. Application of high pressure will expel the molten metal from between the faying surfaces or few indentation marks will be the electrodes on the surfaces.

Hold time

During this time, the increased pressure is maintained on the work pieces. In the beginning of this time, known as forge time, pressure is further increased. The increased pressure is maintained constant afterward.

Advantages

- Similar and dissimilar metals can be welded.
- The process is very fast and it takes very little time to complete the weld.
- No edge preparation is required.
- The process can be used for large production run if multiple spot welding machines are used.

Limitations

- Silver and copper are especially difficult to weld because of their high thermal conductivity
- Welding cannot be performed in all the positions. Only overlapping position is possible.

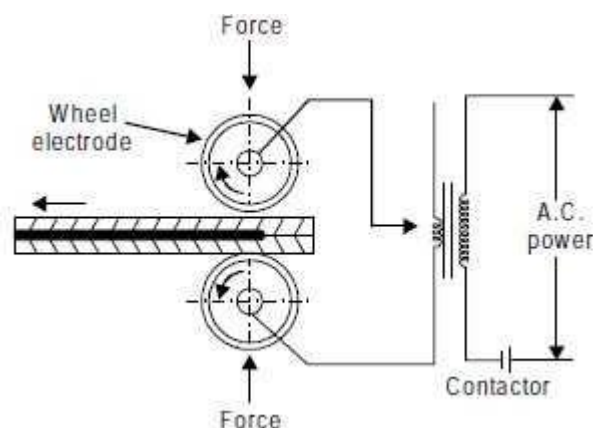
Suitability and Applications

Spot welding can be used as a substitute for riveting. It gives better joint if the work piece thickness does not exceed 3 mm. it finds extensive application in sheet industry. Attachment of handles to domestic cookware and spot welding of automobile bodies are some of its examples.

Resistance seam Welding

Resistance seam welding can be regarded as a continuous spot-welding process. The process uses copper wheels or rollers that act as electrodes in place of cylindrical electrodes for producing the weld. The basic equipment is the same as for spot welding, except for the changes of electrodes.

These electrodes also exert pressure as the sufficient heat is generated at the interface due to the electrical resistance of the parts. The resulting weld consists of a series of overlapping spot weld to provide a pressure- tight .If pressure-tight quality of the lap seam weld is not required, the individual nuggets are spaced to give a stitch effect. This process is called roll spot welding.

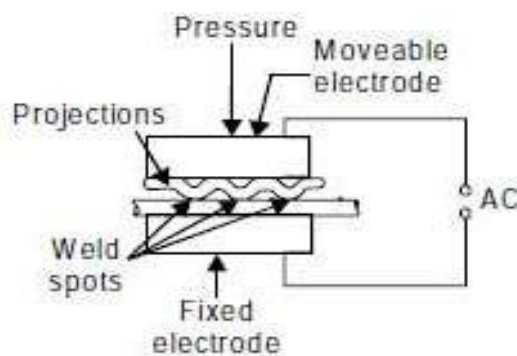


Resistance Projection Welding

Projection welding is a variation of spot welding. Small projections (dimples) are produced on one of the surfaces to be welded with a punch and die at the location where a weld is desired. The workpiece are then placed between large-area electrodes, and the current is switched on. High localized temperatures are generated at the projections.

During the welding process, the projection collapse, owing to heat and pressure, and the workpiece are brought in close contact. Usually two or three projections are welded at a time. Projections may be round, oval or circular to suit the design conditions.

If circular projections are used, their diameter can be equal to the workpiece thickness and extend up to about 60% of the thickness. The weld cycle time for projection welding is the same as for the spot welding. The finished weld is similar to spot weld but has small indentation mark left by the projection.



Advantages

- The process is fast because number of welds can be made simultaneously.
- Small amount of current is required because of greater current density at the projections.
- The pressure to be applied by the electrode is low.
- Thermal shrinkage and distortion are minimum because of low heat affected zone.
- The electrode has longer life. Also, it requires little maintenance.

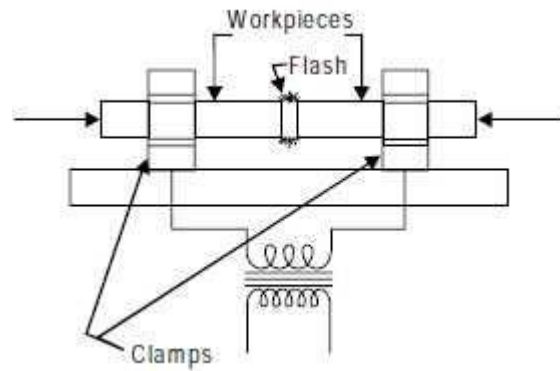
Limitations

Selection of correct current and application of optimum pressure are important.

Flash Welding

Flash welding is also known as flash butt welding. The basic equipment for flash welding is similar to that of butt welding but it differs in the mode of heating the metals. Here the heat required for the joint is obtained by means of an arc rather than the simple resistance heating. So it is classified as arc welding.

A high voltage in the circuit produces a flash (arc) between the parts placed end -to- end. The arc produces enough heat to soften the metal at the joint. The weld is completed by applying suitable amount of forging pressure. The joint may later be machined to remove the upsetting.



Advantages

- It consumes less welding current.
- Larger area can be welded.
- The process is fast because of high heat generation.
- The joint is stronger.

Limitations

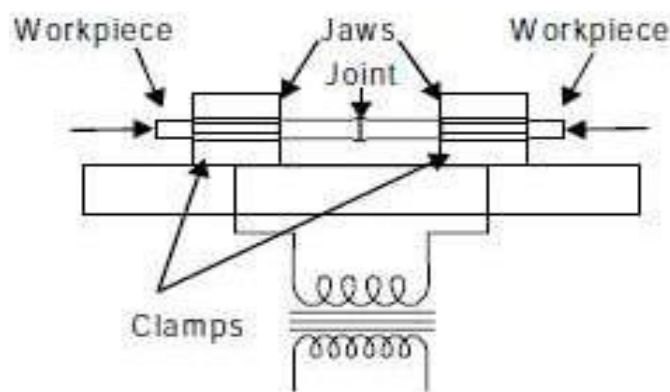
- Metals such as lead, zinc, tin, copper and their alloy are not welded by this method because of their burn-off possibility by the arc.
- Little upsetting can occur at the joint which needs to be machined.
- The welding machine is costly and takes more floor area.

Suitability and Applications

The flash welding is suitable for end- to- end or edge - to – edge joining of similar and dissimilar metal sheets, tubings, bars and rods. Major users are automobile and aircraft industries. Typical application include joining of pipe and tubular shapes for metal furniture and windows.

Resistance Butt Welding

Resistance butt welding is also known as upset welding. In this process, the pieces to be joined are placed end- to-end between two clamps which can exert pressure as per requirement. The high resistance of the joint generates heat and causes fusion to take place at the interface. The joint is now compressed by the clamp force to complete the weld.



Limitations

- Edge preparation is mandatory.
- It requires very high-density current.

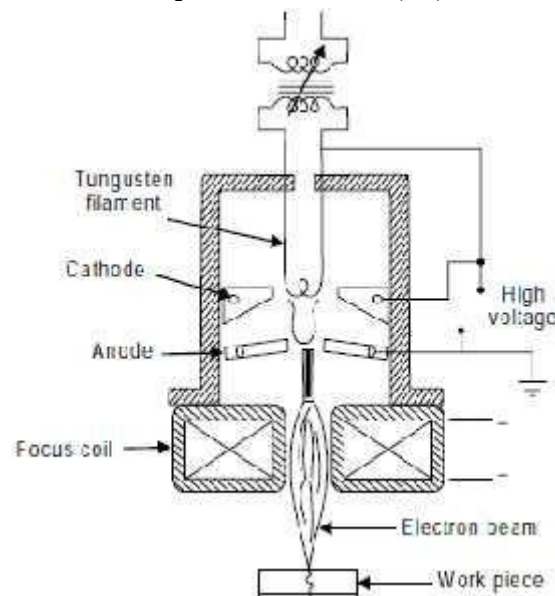
Suitability and Applications

The resistance butt welding is especially suitable for joining rods, pipes and other parts of uniform cross-section.

MODERN WELDING PROCESS

Electron Beam Welding

Electron beam welding is a fusion welding process and has a lot of potentials. In this process heat is generated by a narrow beam of high-velocity electrons. The beam is directed to hit at the desired spot of the workpiece. The kinetic energy of the electrons is converted into heat as they strike the workpiece. The metal is joined by melting the edges of the workpiece or by penetration of the material. The welding operation can be performed in three situations, namely high vacuum, partial vacuum and atmospheric condition (air).



In partial vacuum, welding is performed at low vacuum. In the atmospheric condition welding, the workpiece is placed outside the vacuum chamber. Production rate is increased considerably in this case because of the absence of vacuum. Weld quality is slightly poor but the penetration is greatly reduced.

Advantages

- It produces very high-quality welds.
- Depth-to-width ratio is very high (upto 25:1) because of heat confinement to a narrow area. The intense heat energy can also produce holes in the workpiece (Keyhole Effect).
- There is no requirement of shielding gas, flux or filler metal.
- Higher welding speed is possible. It also ensures low affected zones.
- Distortion and shrinkage in the weld area is minimal.
- Welding can be done in all positions.
- Heat-sensitive materials can be easily welded because of low HAZ.

Limitations

- The equipment cost is very high.
- It is not suitable for large gap joining because of low quantity of molten metal.
- Vacuum is essential to prevent the scattering of electron beam.
- The vacuum requirement reduces the production rate and increases the operational cost.
- Joint preparation is essential to get a better quality weld.

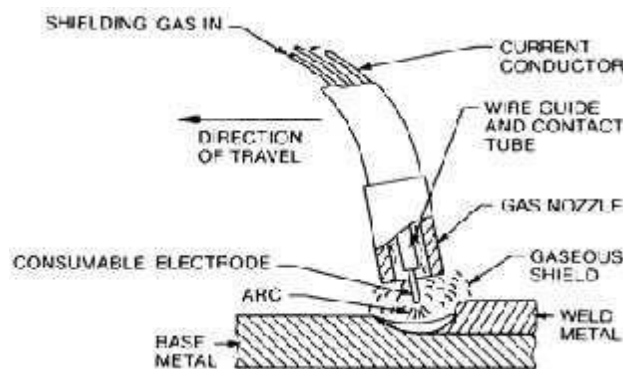
Suitability and Applications

The process is best employed where extremely high quality welds are required. Almost any metal can be welded by this process. The process is used to join common metals, refractory metals, oxidizable metals and various super alloys.

Thinner as well as thicker workpieces can be joined. Workpiece can be a foil or a plate. Typical applications include welding of aircraft, missile, nuclear, and electronic components and gears and shafts in the automotive industry.

Gas metal arc welding process (GMAW)

In this process, an electric arc is struck between a continuously fed consumable electrode wire and the workpiece material as shown. The electrode wire is automatically fed from a spool into the weld pool by a wire feed system. The wire feeder draws the electrode through the welding torch. The shielding is supplied by an inert gas which flows around the wire through a gas cap attached to the torch.



SOLDERING

Welding exceeding 450°C and below the solidus of the base material. Though soldering obtains a good joint between the two plates, the strength of the joint is limited by the strength of the filler metal used.

Solders are essentially alloys of lead and tin. To improve the mechanical properties and temperature resistance, solders are added to other alloying elements such as zinc, cadmium and silver in various proportions. Soldering is normally used for obtaining a neat leak proof joint or a low resistance electrical joint.

The soldered joints are not suitable for high temperature service because of the low melting temperatures of the filler metals used. The soldering joints also need to be cleaned meticulously to provide chemically clean surfaces to obtain a proper bond. Solvent cleaning, acid pickling and even mechanical cleaning are applied before soldering.

To remove the oxides from the joint surfaces and to prevent the filler metal from oxidizing, fluxes are generally used in soldering. Rosin and rosin plus alcohol based fluxes are least active type and are generally used for electrical soldering work. Because of the content of acids, these are corrosive at soldering temperature. is a method of joining similar or dissimilar metals by heating them to a suitable temperature and by means of a filler metal, called solder, having liquid us temperuatre.

Solders

Solders are alloys of lead and tin. Solder may also contain certain other elements like cadmium, and antimony in small quantities. The percentage composition of tin and lead determines the physical and mechanical properties of the solder and the joint made. Most solder is available in many forms-bar, stick, fill, wire, strip, and so on. It can be obtained in circular or semi-circular rings or any other desired shape. Sometimes the flux is included with the solder. For example, a cored solder wire is a tube of solder filled with flux.

Solder Fluxes

The flux does not constitute a part of the soldered joint. Zinc chloride, ammonium chloride, and hydrochloric acid are the examples of fluxes commonly used in soldering. The function of fluxes in soldering is to remove oxides and other surface compounds from the surfaces to be soldered by displacing or dissolving them. Soldering fluxes may be classified into four groups-

- (1) Inorganic fluxes (most active)
- (2) Organic fluxes (moderately active)
- (3) Rosin fluxes (least active), and
- (4) Special fluxes for specific applications

BRAZING

Like soldering, brazing is a process of joining metals without melting the base metal. Filler material used for brazing has liquidus temperature above 450°C and below the solidus The filler metal is drawn into the joint by means of capillary action (entering of fluid into tightly fitted surfaces). Brazing is a much widely used joining process in various industries because of its many advantages. Due to the higher melting point of the filler material, the joint strength is more than in soldering

Methods of Brazing

Torch Brazing

It is the most widely used brazing method. Heat is produced, generally, by burning a mixture of oxy-acetylene gas, as in the gas welding. A carbonizing flame is suitable for this purpose as it produces sufficiently high temperature needed for brazing.

Furnace Brazing

It is suitable for brazing large number of small or medium parts. Usually brazing filler metal in the granular or powder form or as strips is placed at the joint, and then the assembly is placed in the furnace and heated. Large number of small parts can be accommodated in a furnace and simultaneously brazed.

Braze Welding

In welding processes where the joint of the base metal is melted and a joint is prepared having higher joint strength, it is likely to cause metallurgical damage by way of phase transformations and oxide formation. In this process, the base metal is not melted, but the joint is obtained by means of a filler metal.